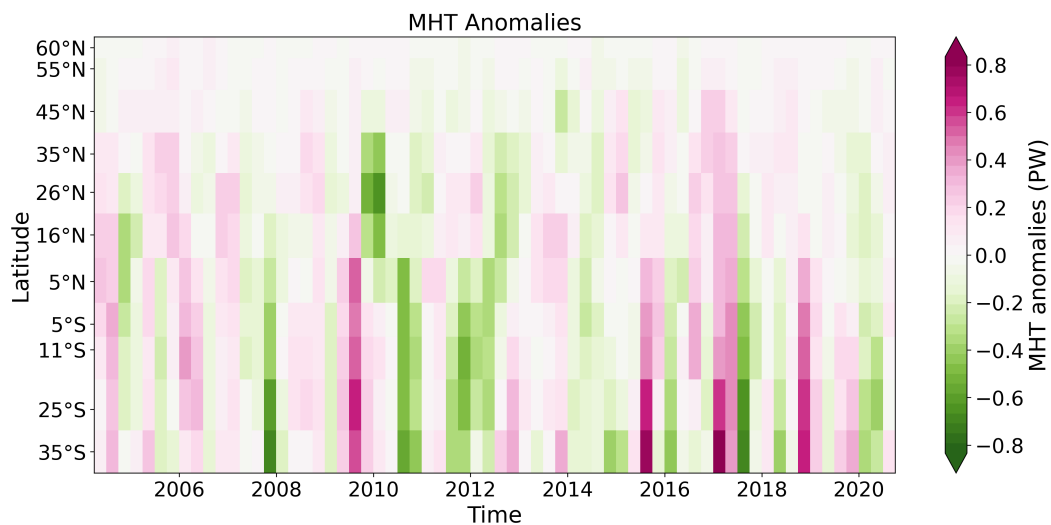


Masterthesis

1. Concept - Abstract

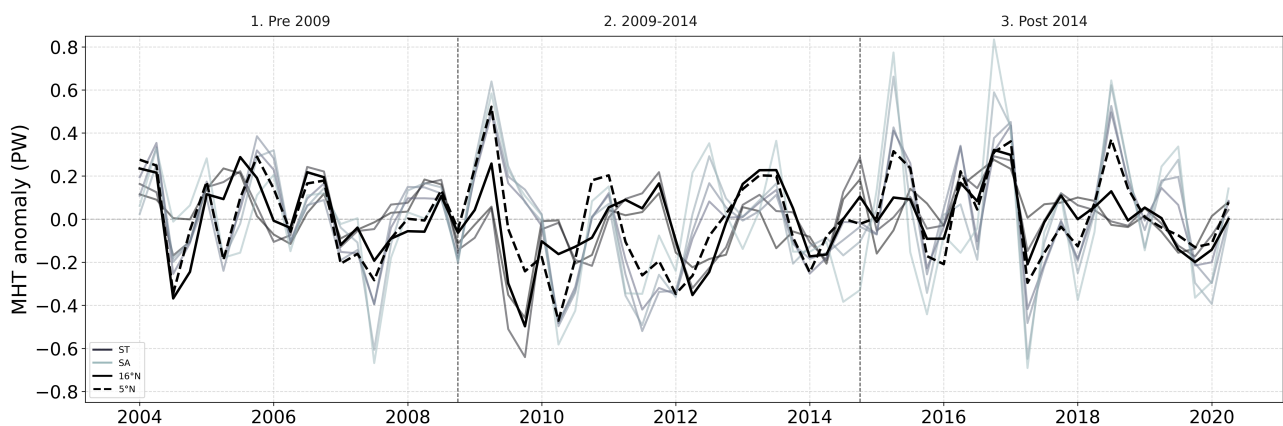
1.1 Result outline:

1. The hovmöller diagram of the meridional heat transport anomalies hint towards latitudes where anomalies vary together, namely between 35°N - 16°N and 5°N - 35°S . Overall weakest anomalies north of 35°N , in subpolar region.

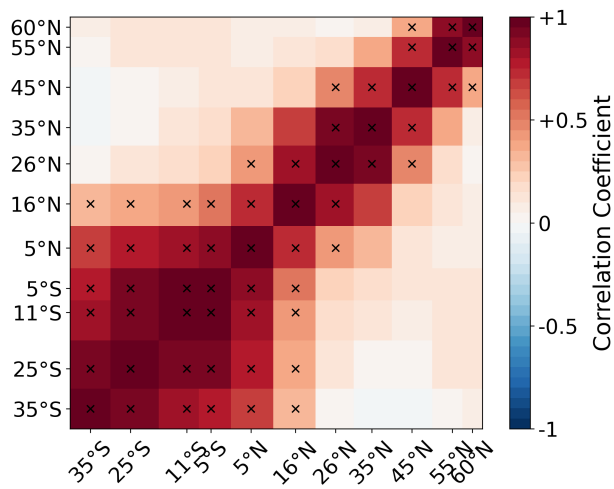


2.

(At the beginning of the 2nd period the subtropical north atlantic latitudes anomalies show the opposite behavior of the south atlantic ones.) ? show the timeseries but unsure about what to say or when ?

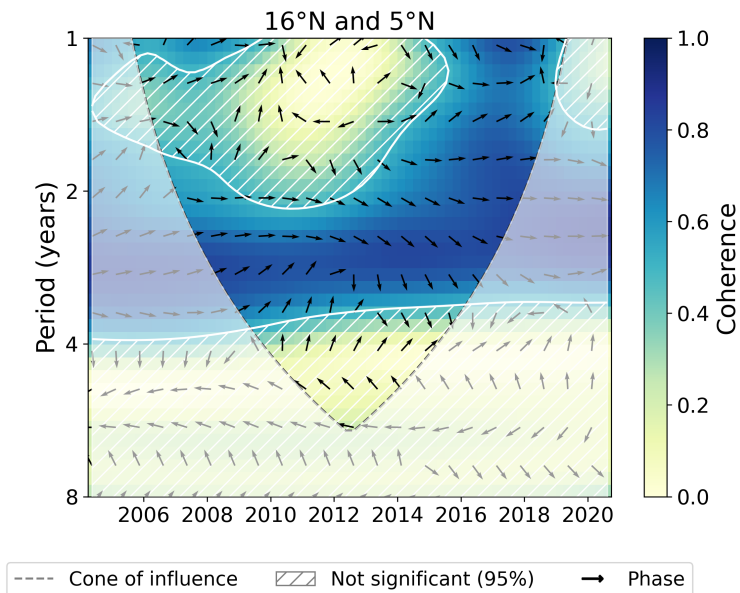


3. The cross-correlation between all latitudes confirms that these latitude bands tend to vary together, which can be seen from the high pairwise correlation between those.



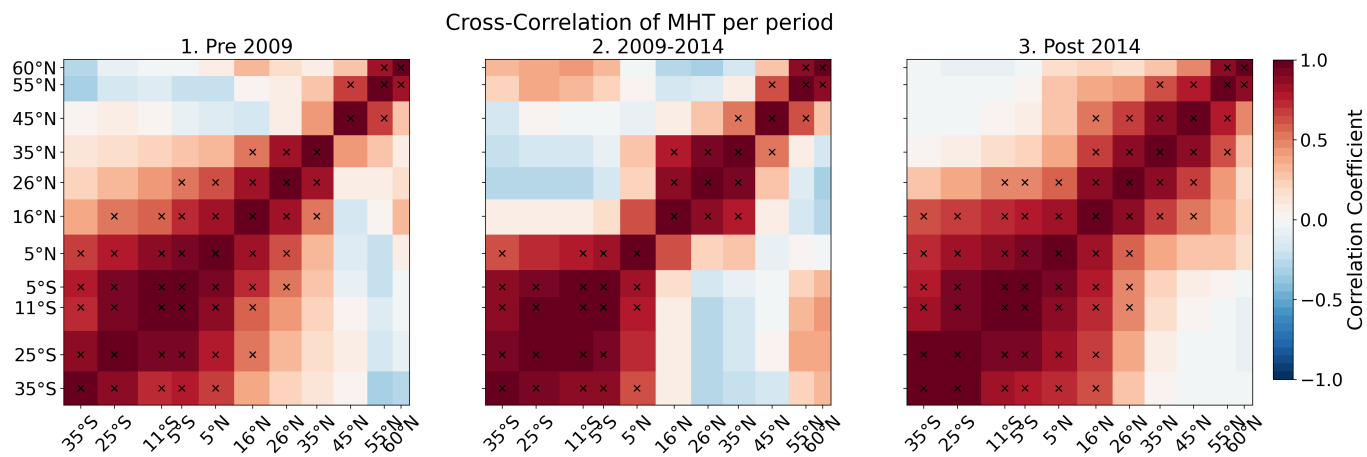
4. The wavelet coherence between 16 and 5°N reveals a drop in the coherence of the meridional heat transport occurring between 2009-2014.

4.1 Three periods can be distinguished: 2004-2008, 2009-2014, 2015-2020



5. MAIN FIGURE

The cross correlation done in 1. Redone for the 3 periods reveals that prior to 2009 and after 2014 the meridional heat transport seems coherent throughout the atlantic south of 35°N, while in 2009-2015 the coherence splits, leaving two boxes of coherent MHT.



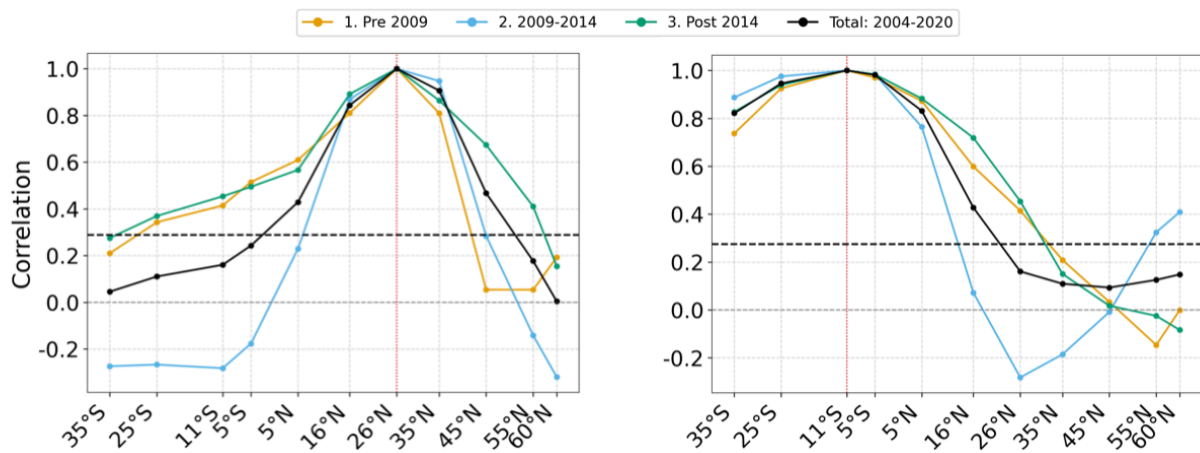
6. The average correlation between the two blocks drops for the indicated period from 0.53 to a no longer significant value.

	A	B	C
1	Period	Corr ST & SA block	Significant (95%)
2	Pre 2009	0.53	Yes
3	2009-2014	0.02	No
4	Post 2014	0.47	No (rcrit: 0.48)

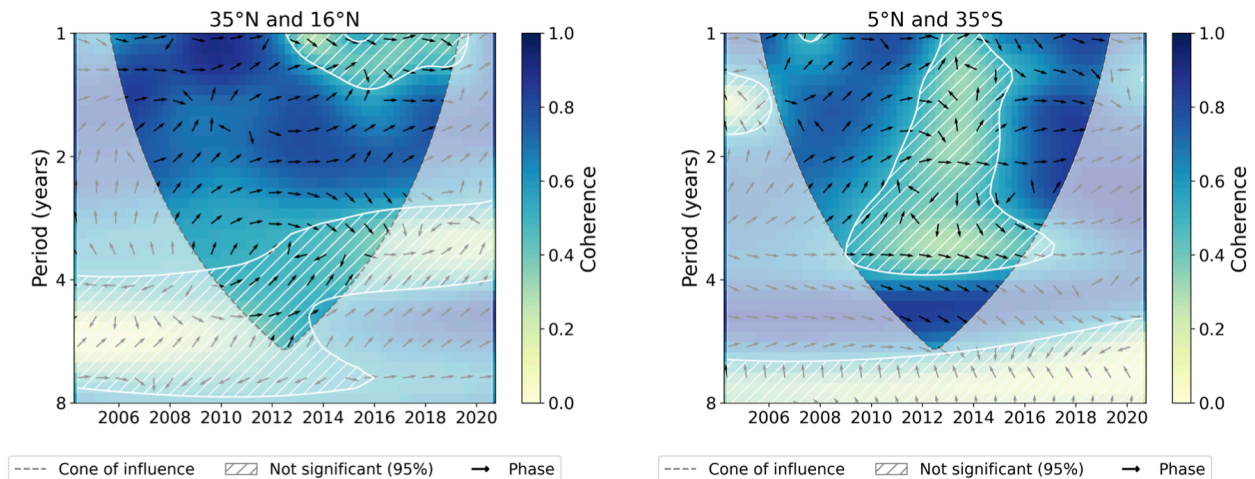
7. The blocks in itself stay coherent throughout the whole timeseries, with just slight changes in the connection.

(can show this either with 0-lag correlation or wavelet coherence)

0-lag again 26°N and 11°S

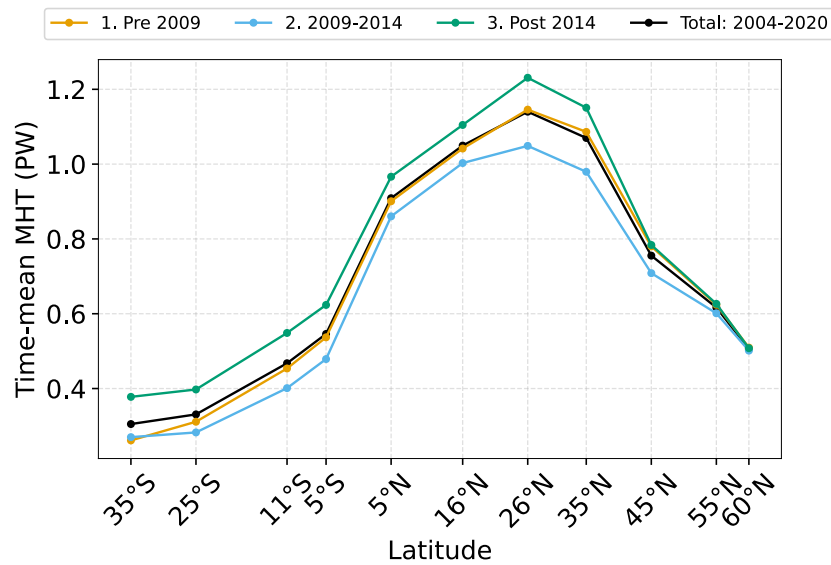


Or WCT: Subtropical north atlantic & south atlantic latitudes:



8. Compared to the pre2009 baseline period, the MHT reduces at all latitudes, with the strongest reduction at 26°N. After 2014 the MHT recovers, while even surpassing the pre2009 baseline at all latitudes.

— here: add 95% confidence intervals!



9. heat budget decomposition

the reduction in MHT is due to ? Compared to the pre 2009 baseline the air sea heat flux in the region 16°N - 5°N increases around 15% while the heat transport convergence decreases by only 1.5%.

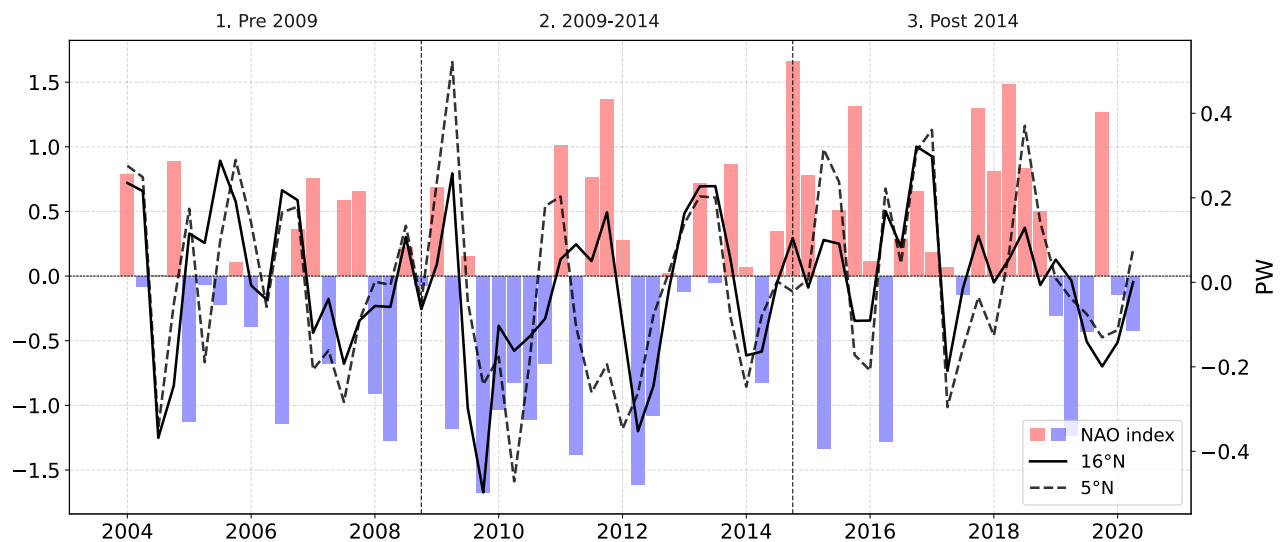
Maybe table for this

10. I. Found high correlation of the subtropical north atlantic heat transport with the NAO index for the period 2009-2014 of around 0.50 where the strong negative anomalies in MHT correlate with a negative NAO event in winter of 2009 and 2010, compared to the pre2009 baseline (-0.2),

Connection to coherence split?

(Plot: MHT anomalies, not detrended vs. NAO index)

South of 5°N there is no such correlation visible. Disconnected ? Gyre circulation (discussion)



1.7 concept abstract

(Background) The Atlantic ocean is one of the dominant heat sinks for excess heat in the climate system, with a high uptake and redistribution driven by the Atlantic Meridional Overturning Circulation (AMOC). Understanding how the Atlantic meridional heat transport (MHT) varies across latitudes and also connects with the AMOC is essential.

(Question) However, due to sparse observations the basin wide variability of MHT remains poorly understood, which leads me to investigate whether variations of MHT propagate coherent throughout the Atlantic basin, and if there are specific events which can trigger alterations.

(Exp approach/Methods) Using satellite-based estimates from Calafat et al. (2025) I evaluated the coherence of Atlantic MHT at 12 latitude bands between 2004-2020, spanning from 65°N to 35°S. I used statistical methods such as correlation and wavelet coherence and connecting MHT anomalies to the NAO index.

(Result) I found that in 2009 the relationship between the subtropical north Atlantic latitudes and the south Atlantic latitudes changed, leading to 2 disconnected (*covarying*) boxes, with a coherence drop specifically in the region of 16°N - 5°N, whereas before the heat transport was connected throughout.

However, after 2014 a recovery in the connection is found, resulting in an enhancement of the coherence as well as the mean heat transport values, which are both superior to its previous state. In the weaker period (2009-2014) MHT anomalies were more strongly correlated with the NAO index, than before and after.

(Answer / Implication) These results indicate that an event such as the record low NAO in 2009 can affect not only the overall heat transport across the Atlantic but can change the coherence pattern locally. However, the results also show that the heat transport, even if its change is persistent for several years, is able to restore its pre-break state afterwards, even exceeding it.